
On the hardness of proving a ZX-Diagram equality

— Mathis Hage —

Summary

- I. Quantum computing in 1 minute
- II. The ZX-Calculus language
- III. Circuit Obfuscation
- IV. Complexity of proving a diagram equality
 - A. General case (rational phases)
 - B. Extension to Real Phases
 - C. Clifford Fragment : (hardly) easier ?
- V. Conclusion

I. Quantum Computing in 1 minute

I. Quantum Computing in 1 minute

~~Classical bits~~ → Qubits : $\alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

I. QC in 1 min.: Measuring a Qubit

i) Initial Qubit

$$\alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

ii) Measurement



iii) Obtained Result

$$\begin{cases} |0\rangle, & \text{with probability } |\alpha|^2 \\ |1\rangle, & \text{with probability } |\beta|^2 \end{cases}$$

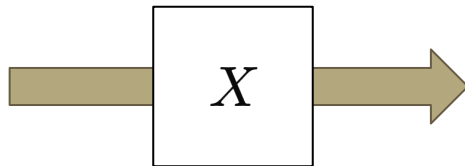
I. QC in 1 min. : Modifying a Qubit

Manipulating Qubits : Gates

i) Initial Qubit

$$\alpha|0\rangle + \beta|1\rangle$$

ii) Goes through gate "X"



iii) final state

$$\beta|0\rangle + \alpha|1\rangle$$

I. QC in 1 min. : Modifying a Qubit

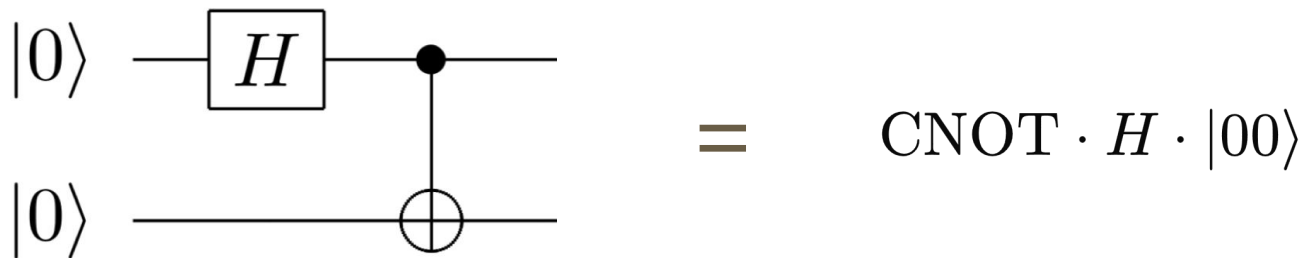
Going Through a Gate = Matrix Multiplication !

“Qubit goes through gate X” means :

$$X(\alpha|0\rangle + \beta|1\rangle) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \beta|0\rangle + \alpha|1\rangle$$

I. QC in 1 min. : Circuits

Quantum Circuit Notation = Apply Gates From Left to Right



The Things To Remember

- A Qubit is a vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{C}^2$
- To manipulate a qubit : apply gates = multiply the vector by a matrix
- To represent these matrices : quantum circuit notation

II. The ZX-Calculus Language

Purpose

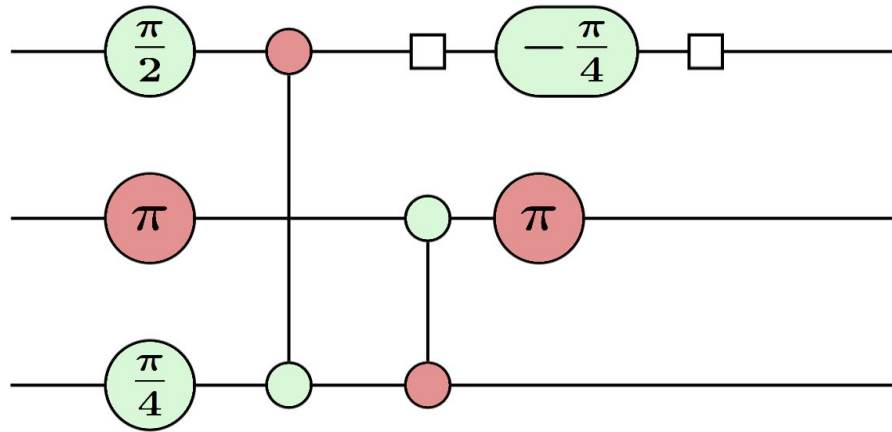
Diagrammatic language to manipulate linear maps on Qubits

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A way to represent and reason on the matrices $U \in \mathbb{C}^{2^m \times 2^n}$

Example

Close to Quantum Circuit notation :



Generators : Spiders

Z-spider
(n inputs, m outputs)

$$n \left\{ \begin{array}{c} \text{diagram of Z-spider with } n \text{ inputs and } m \text{ outputs, center labeled } \alpha \end{array} \right\} m = \overbrace{|0 \dots 0\rangle}^m \overbrace{\langle 0 \dots 0|}^n + e^{i\alpha} \overbrace{|1 \dots 1\rangle}^m \overbrace{\langle 1 \dots 1|}^n$$

X-spider
(n inputs, m outputs)

$$n \left\{ \begin{array}{c} \text{diagram of X-spider with } n \text{ inputs and } m \text{ outputs, center labeled } \alpha \end{array} \right\} m = \overbrace{|+\dots+\rangle}^m \overbrace{\langle +\dots+|}^n + e^{i\alpha} \overbrace{|-\dots-\rangle}^m \overbrace{\langle -\dots-|}^n$$

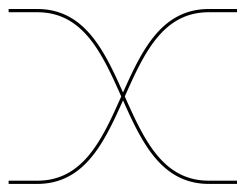
Other Generators



= Bell state $|00\rangle + |11\rangle$



= Bell Effect $\langle 00| + \langle 11|$



= SWAP $|00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|$

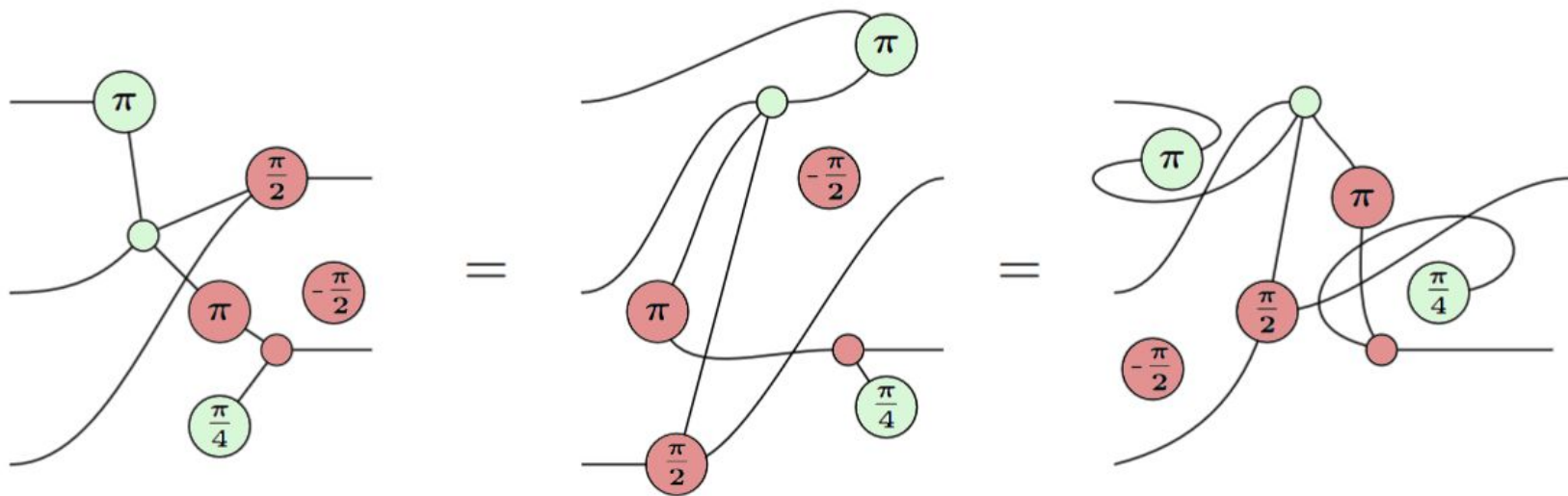
Universality

Claim :

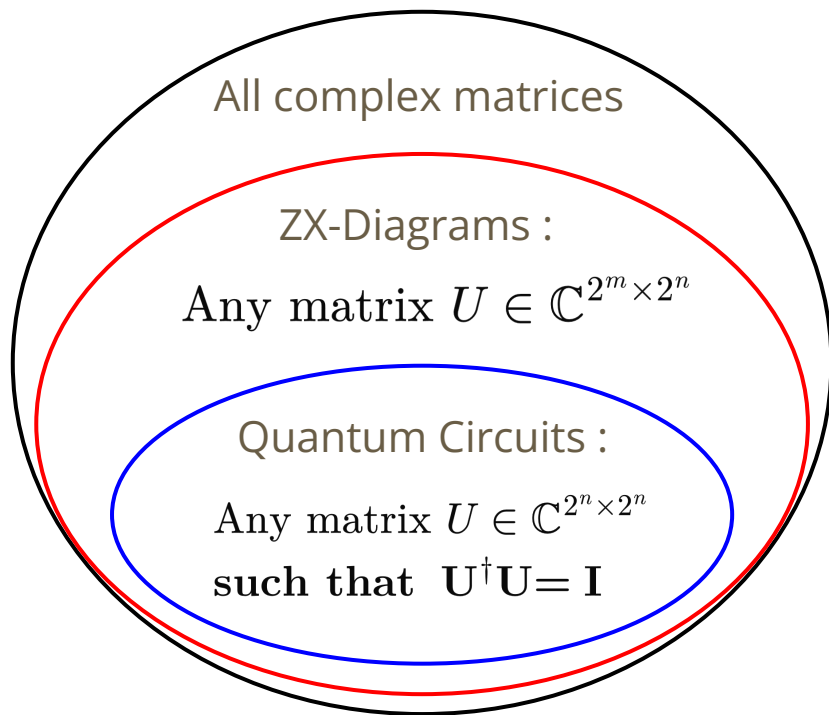
By arranging the generators in all sorts of ways...

... We can build any matrix $U \in \mathbb{C}^{2^m \times 2^n}$

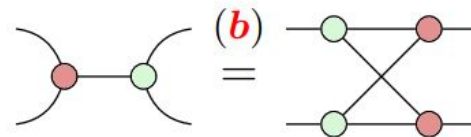
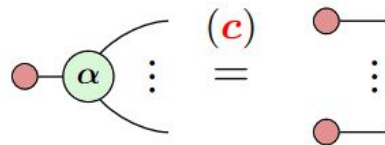
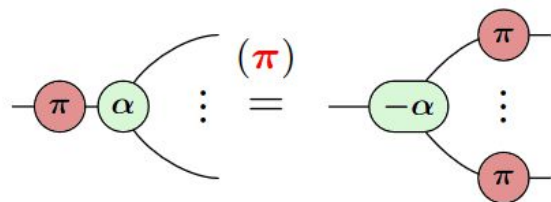
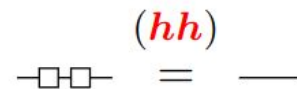
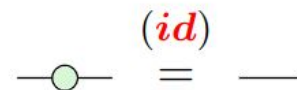
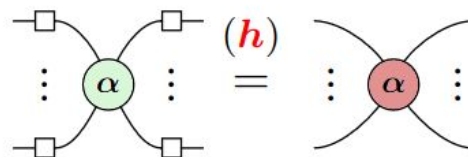
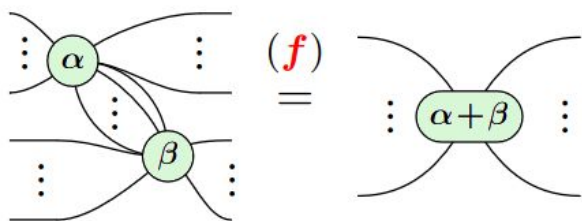
Only Connectivity matters



ZX-Diagrams $\neq$ Quantum Circuits



Rule Set



Completeness

ZX-calculus is complete

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Two ZX-diagrams are equal **if and only if** we can go from one to the other using only operations from the allowed rules.

The Things To Remember

- A ZX-diagram encodes a specific linear map (a matrix)
- A ruleset is given ; applying a rule to a ZX-diagram doesn't change the matrix it represents
- Two diagrams represent the same matrix **if and only if** we can go from one to the other using only rules from the ruleset.

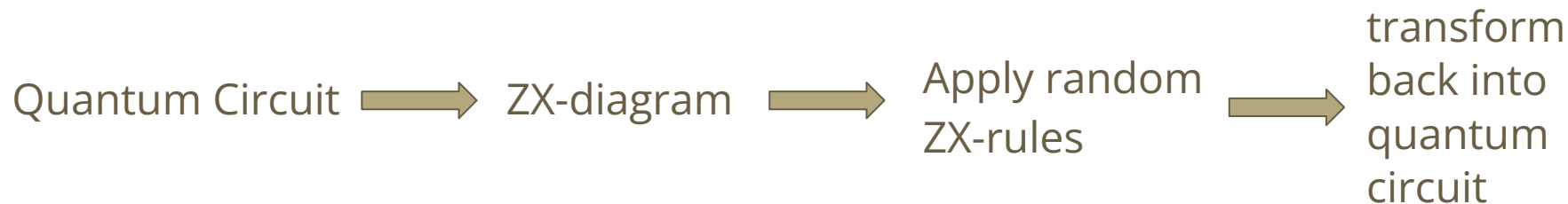
III. Circuit Obfuscation

The Idea

Start from a random seed. Design an algorithm to generate a quantum circuit from the seed.

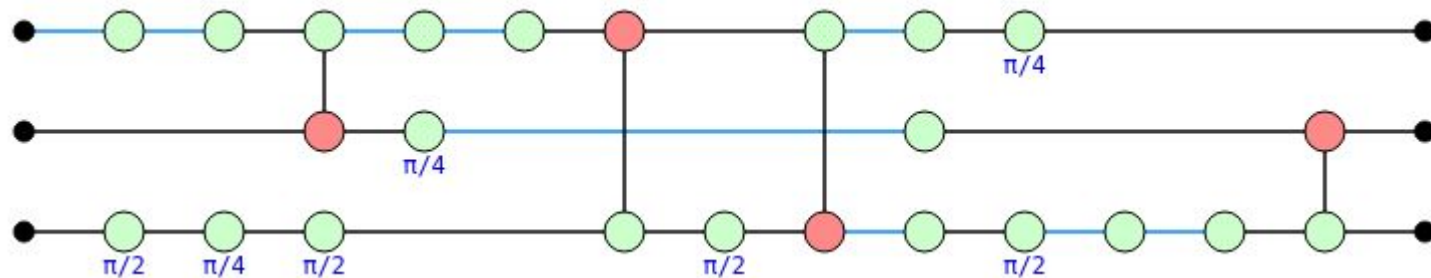
Goal : make it hard for an attacker to recover the seed when given the quantum circuit you generated

In practice



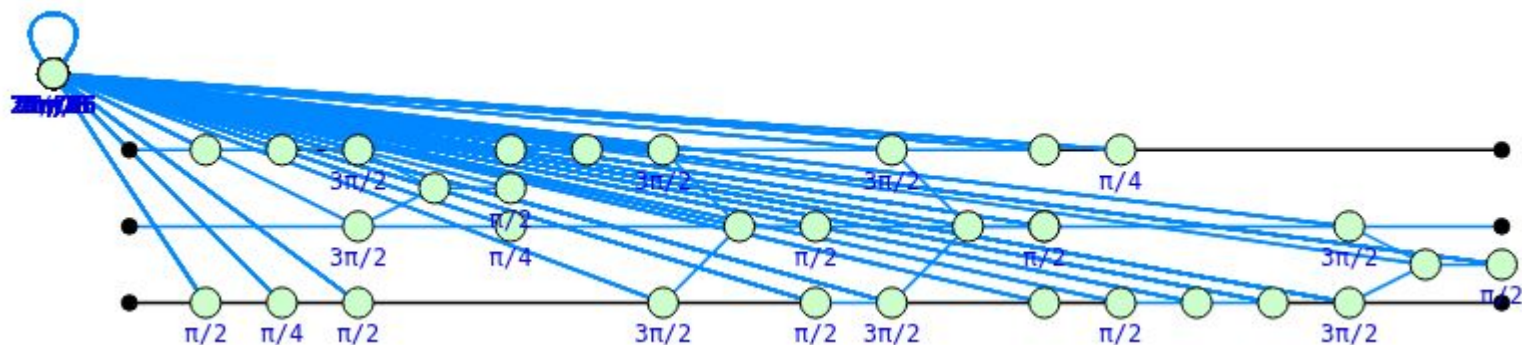
Example

Initial circuit (converted to ZX-diagram) :



Example

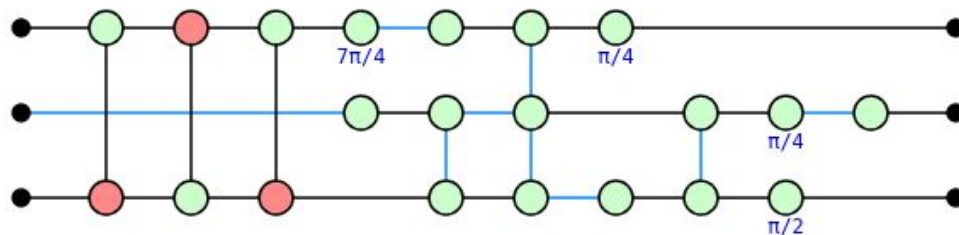
Use the rules to convert all the spiders to green ones and add identities randomly :



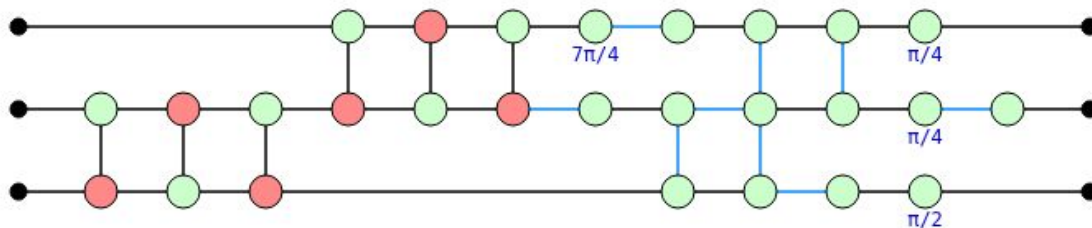
Example

Extract a circuit using standard algorithms.

Original circuit :



Obfuscated circuit :



IV. Complexity of Proving a Diagram Equality

The problem

Given Two ZX-Diagrams **representing the same tensor...**

... How hard is it to prove the equality **using only the rules of ZX-Calculus ?**

The problem

In other words, how can we classify the language :

$\text{ZX-REWRITE} := \{(D_1, D_2, k) \mid D_1, D_2 \text{ are ZX-diagrams s.t. the equality } D_1 = D_2 \text{ can be proven true in at most } k \text{ steps}\}.$

A. Rational Phases, $\alpha \in \mathbb{Q}$

1. NP-Membership

Construct a polytime verifier :

On input : $\langle D_1, D_2, k, C \rangle$,

Parse C as $\langle R, L \rangle = \langle (r_1, \dots, r_n), (l_1, \dots, l_n) \rangle$

If $n > k$, reject.

Otherwise, apply the rules $r_1, \dots, r_n$ sequentially to D_1

Accept if and only if D_1 and D_2 are the same.

A. Rational Phases, $\alpha \in \mathbb{Q}$

Ensuring that the verifier works :

- Phases can be passed as inputs (finite presentation)
- No rule makes the number of nodes explode exponentially
- Possible to adopt a naming convention for the nodes / wires to ensure consistency between the verifier and the certificate

A. Rational Phases, $\alpha \in \mathbb{Q}$

2. NP-hardness

Idea for a reduction :

Hardness of the problem may come from the decision of how to split a phase γ into $\gamma = \alpha + \beta$

A. Rational Phases, $\alpha \in \mathbb{Q}$

2. NP-hardness

Reduction to a simpler language :

$$\text{ZX-REWRITE}_F := \{ \langle D_1, D_2, k \rangle \in \text{ZX-REWRITE} \mid \text{we can go from } D_1 \text{ to } D_2 \text{ using only the fusion rule, at most } k \text{ times} \}.$$

A. Rational Phases, $\alpha \in \mathbb{Q}$

2. NP-hardness

Claim :

$$\text{SUBSET-SUM} \leq_p \text{ZX-REWRITE}_F$$

A. Rational Phases, $\alpha \in \mathbb{Q}$

2. NP-hardness

Where a SUBSET-SUM instance is the 2-tuple $\langle S, t \rangle$ with :

- $S = \{s_1, \dots, s_n\} \subset \mathbb{N}$
- $t \in \mathbb{N}$

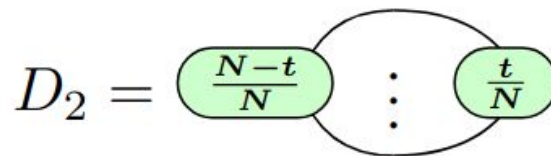
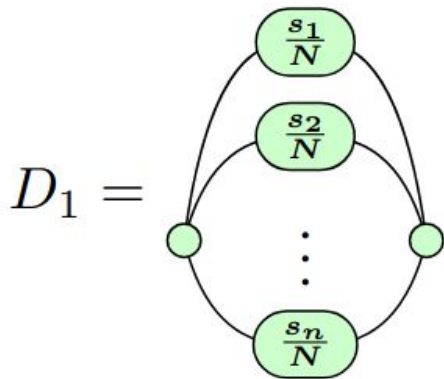
Such that :

- $\sum_{s \in S} s = t$

A. Rational Phases, $\alpha \in \mathbb{Q}$

Reduction function $f :=$

On input $\langle S, t \rangle$, set $N \leftarrow \sum_{s \in S} s$, and build :



Return $\langle D_1, D_2, n \rangle$

A. Rational Phases, $\alpha \in \mathbb{Q}$

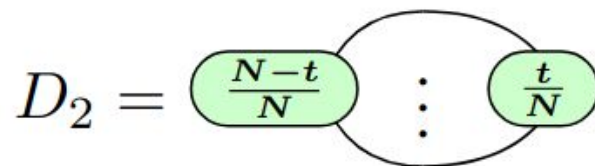
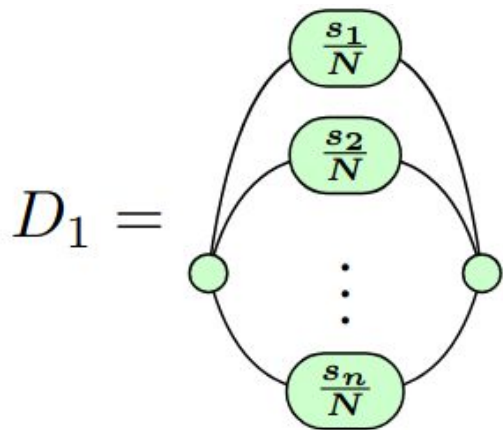
Reduction correctness:

We show that :

$$\langle S, t \rangle \in \text{SUBSET-SUM} \Leftrightarrow \langle D_1, D_2, n \rangle \in \text{ZX-REWRITE}_F$$

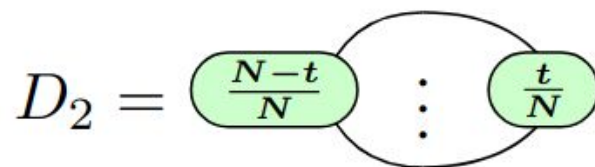
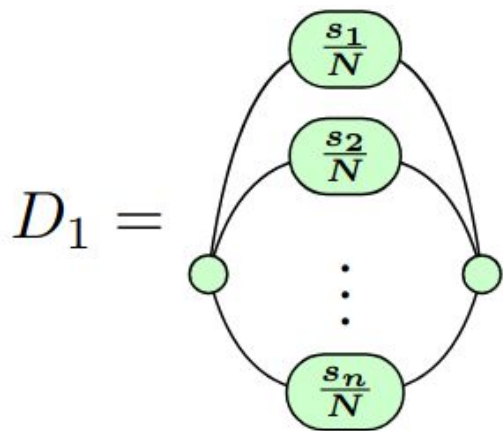
A. Rational Phases, $\alpha \in \mathbb{Q}$

Reduction correctness: $(\Rightarrow)$ Suppose $\langle S, t \rangle \in \text{SUBSET-SUM}$:



A. Rational Phases, $\alpha \in \mathbb{Q}$

Reduction correctness: ($\Leftarrow$) Suppose $\langle D_1, D_2, n \rangle \in \text{ZX-REWRITE}_F$:



A. Rational Phases, $\alpha \in \mathbb{Q}$

Extension of the result to the initial problem with all rules, ZX-REWRITE :

Only problem is to adapt $(\Leftarrow)$:

We argue that using any other rule when trying to go from D_1 to D_2 in at most n steps is impossible.

B. Extension To Real Phases, $\alpha \in \mathbb{R}$

Framework of the **Blum-Shub-Smale (BSS)** machines.

Definition (linear BSS machine) :

- Usual tape of Turing machines replaced by a vector $x \in \mathbb{R}^\infty$
- Possible operations on components of x :

$$(\cdot, +, -, =)$$

- We denote the complexity classes of such machines as $\mathcal{C}_{\text{lin}}^=$

B. Extension To Real Phases, $\alpha \in \mathbb{R}$

Theorem :

Let $BP(\mathcal{C}) := \{L \cap \{0, 1\}^* \mid L \in \mathcal{C}\}$, then :

$$\text{NP} = BP(\text{NP}_{\text{lin}}^{\overline{}})$$

$$\text{P} = BP(\text{P}_{\text{lin}}^{\overline{}})$$

B. Extension To Real Phases, $\alpha \in \mathbb{R}$

Claim(s) :

- ZX-REWRITE (real phases) is in $\text{NP}_{\text{lin}}^{\overline{\overline{}}}$
- ZX-REWRITE_F (real phases) is $\text{NP}_{\text{lin}}^{\overline{\overline{}}}$ -hard
- We argue in a similar fashion that ZX-REWRITE (real phases) is $\text{NP}_{\text{lin}}^{\overline{\overline{}}}$ -hard

B. Extension To Real Phases, $\alpha \in \mathbb{R}$

However, in this case the claims have a different interpretation, as it is known that :

$$P_{\text{lin}}^{\overline{=}} \neq NP_{\text{lin}}^{\overline{=}}$$

C. Clifford Fragment : (hardly) easier ?

Restrict diagrams to the *Clifford fragment*, meaning :

$$\alpha \in \{0, \pi/2, \pi, 3\pi/2\}$$

Intuitively, we expect the problem to be easier

C. Clifford Fragment : (hardly) easier ?

Moreover : recall the idea for the previous reduction :

~~Hardness of the problem may come from the
decision of how to split a phase γ into $\gamma = \alpha + \beta$~~

We can check all possible splits of a spider in constant time...

C. Clifford Fragment : (hardly) easier ?

Problem similarly shown to be in NP (same verifier), but for hardness...

...Did not solve due to a lack of time, however :

Conjecture :

ZX-REWRITE (Clifford fragment) is NP-hard

V. Conclusion

References

- "Measurement operator" image, slide 5 :
<https://tex.stackexchange.com/questions/716484/how-to-draw-a-quantum-measurement-in-tikz-latex>
- Quantum Circuit image, slide 8 : [https://dojo.qulacs.org/en/latest/notebooks/1.4 quantum circuit diagram.html](https://dojo.qulacs.org/en/latest/notebooks/1.4_quantum_circuit_diagram.html)
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